

HALLANDS VADERO, Sweden

WMO: 026050

Lat: 56.4496N

Lon: 12.5453E

Elev: 10

StdP: 101.20

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-5.6	-11.9	1.4	-4.8	-9.4	1.7	-4.5	18.0	5.1	16.3	4.9	5.1	60	0.709

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.8	24.4	18.9	22.9	18.4	21.6	17.7	20.0	23.0	19.2	21.9	18.3	20.9	4.3	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	21.5	18.0	13.0	20.8	17.2	12.3	20.0	57.2	23.1	54.6	22.0	51.9	21.0	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.1	13.5	12.3	DB	-8.2	27.3	2.4	1.6	-9.9	28.5	-11.4	29.4	-12.7	30.3	-14.5	31.5	
			WB	-9.0	21.2	2.2	0.9	-10.6	21.8	-11.9	22.3	-13.2	22.8	-14.8	23.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.1	0.9	1.0	3.0	7.1	12.1	15.4	18.1	17.8	14.6	10.2	5.9	2.8
	DBStd	6.77	3.25	2.95	2.69	2.84	2.88	2.22	2.25	1.98	2.18	2.58	2.98	3.47
	HDD10.0	1238	283	251	218	96	9	0	0	0	0	30	125	224
	HDD18.3	3415	542	484	476	337	195	92	33	34	112	253	374	483
	CDD10.0	913	0	0	0	9	74	162	249	242	139	35	2	0
	CDD18.3	49	0	0	0	0	2	4	24	18	2	0	0	0
	CDH23.3	102	0	0	0	1	9	7	54	30	1	0	0	0
	CDH26.7	8	0	0	0	0	2	1	5	1	0	0	0	0
Wind	WSAvg	6.1	6.5	6.5	5.8	5.2	5.2	5.8	5.4	5.9	6.4	6.8	6.9	7.0
Precipitation	PrecAvg	725	55	45	37	36	50	72	81	90	67	68	59	61
	PrecMax	926	118	98	77	80	125	129	210	198	128	120	115	113
	PrecMin	528	5	6	10	6	17	6	2	28	15	12	13	14
	PrecStd	100	27	19	18	20	25	34	45	39	31	28	26	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.7	6.6	11.5	18.7	24.3	24.2	27.1	25.8	22.6	16.4	12.2	9.5
		MCWB	6.9	5.3	7.7	12.6	16.6	18.0	20.2	19.3	17.5	14.0	11.5	8.7
	2%	DB	6.7	5.7	8.3	15.4	20.4	22.3	24.8	24.0	20.4	15.0	11.2	8.5
		MCWB	5.8	4.8	6.1	10.5	14.8	17.3	19.7	18.6	16.5	13.3	10.4	7.7
	5%	DB	5.7	5.2	6.9	13.1	18.2	20.6	23.2	22.5	18.7	14.2	10.5	7.7
		MCWB	4.9	4.4	5.4	9.4	13.9	16.5	19.0	18.3	15.7	12.7	9.6	7.0
	10%	DB	4.9	4.6	6.0	11.1	16.3	19.0	21.8	21.0	17.6	13.4	9.7	7.0
		MCWB	4.3	3.9	4.9	8.4	13.0	15.7	18.2	17.7	15.1	12.1	8.7	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	5.7	8.0	13.7	17.6	19.7	21.4	20.8	18.6	14.7	11.7	8.7
		MCDB	7.1	6.1	10.9	18.3	22.8	22.8	25.2	23.5	21.1	15.7	12.0	8.9
	2%	WB	5.6	5.1	6.5	11.2	15.8	18.3	20.3	19.7	17.2	13.7	10.7	7.7
		MCDB	6.0	5.5	7.9	14.1	19.0	21.2	23.6	22.4	19.4	14.6	11.0	8.0
	5%	WB	4.9	4.6	5.7	9.9	14.7	17.1	19.5	18.9	16.3	13.0	9.8	6.9
		MCDB	5.4	5.0	6.6	12.3	17.5	19.6	22.5	21.3	18.2	13.9	10.4	7.3
	10%	WB	4.3	4.0	5.1	8.8	13.5	16.2	18.7	18.1	15.5	12.3	8.9	6.2
		MCDB	4.8	4.6	5.8	10.7	15.8	18.6	21.3	20.5	17.3	13.3	9.6	6.7
Mean Daily Temperature Range	5% DB	MDBR	2.7	2.7	3.7	5.2	5.5	4.9	4.8	4.7	4.2	3.2	2.6	2.6
		MCDBR	2.3	2.5	4.9	8.6	9.1	7.7	7.5	7.5	6.0	3.7	2.6	2.3
		MCWBR	2.6	2.2	3.5	5.4	5.6	5.0	4.5	4.4	3.8	3.0	2.8	2.6
	5% WB	MCDBR	2.4	2.4	4.4	7.4	7.7	6.7	6.7	6.0	5.3	3.4	2.6	2.4
		MCWBR	2.6	2.2	3.3	4.8	5.1	4.6	4.3	4.2	3.8	3.1	2.8	2.7
Clear-Sky Solar Irradiance	taub	0.310	0.334	0.354	0.389	0.385	0.377	0.400	0.401	0.370	0.354	0.333	0.300	
	taud	2.275	2.307	2.329	2.294	2.376	2.425	2.376	2.380	2.445	2.403	2.321	2.249	
	Ebn at Noon	577	719	803	826	854	866	834	807	783	689	544	483	
	Edh at Noon	53	75	94	113	110	107	110	103	84	69	51	42	
All-Sky Solar Radiation	RadAvg	0.41	1.00	2.35	3.96	5.17	5.43	5.04	4.05	2.85	1.37	0.51	0.28	
	RadStd	0.08	0.16	0.24	0.53	0.49	0.47	0.60	0.35	0.33	0.21	0.09	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	-120	-172	N/A	N/A

Nomenclature: See separate page